

MechaScreener: Large Language Model-Based Automated Screening for Systematic Reviews and Research

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Abstract

Systematic Reviews (SRs) are the gold standard for evidence synthesis, but the manual title and abstract screening of thousands of references creates a severe bottleneck. While automated tools exist, they have historically struggled to achieve the near-perfect recall required for safe evidence synthesis. This paper presents MechaScreener, a “zero-shot” automated screening tool that utilises a Large Language Model (LLM) to rank article relevance. Because it is zero-shot, the tool requires no initial training data or manual pre-screening. Instead, MechaScreener directly applies a user-provided PICOT or general eligibility criteria to assign a deterministic inclusion probability score (1–5) to each reference. We evaluated the tool in two phases: a development phase using five reference libraries to optimise prompts, and an independent evaluation phase across 10 diverse Cochrane review libraries (comprising both randomised controlled trials and non-RCTs) containing over 58,000 references. In the evaluation dataset, MechaScreener achieved a perfect mean recall of 1.00 (100%), ensuring no relevant articles were missed. Concurrently, it achieved an overall mean specificity of 0.61 (61%). Specificity varied by domain—ranging from 0.21 in broad public health topics to 0.91 in precise pharmacological interventions—reflecting the tool’s built-in conservatism when evaluating ambiguous abstracts. By safely eliminating over 60% of irrelevant literature during the initial screening phase without compromising recall, MechaScreener functions as a highly reliable “first-pass” filter, allowing researchers to substantially reduce manual workloads and reallocate resources toward full-text review and data extraction.

Highlights

What is already known

- Systematic reviews are time-consuming and resource-intensive, with manual title and abstract screening acting as the primary bottleneck.
- Existing automation tools and Large Language Models (LLMs) show promise but historically struggle to achieve the near-perfect recall required for safe evidence synthesis.

What is new

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- MechaScreener is a zero-shot, LLM-based screening tool that utilizes a deterministic 1–5 probabilistic scoring mechanism based on PICOT criteria.
- Across 15 systematic review datasets, MechaScreener achieved a perfect mean recall (100%) while safely reducing the manual screening workload by an average of 61%.

Potential impact for Research Synthesis Methods readers

- Researchers can deploy this tool as a highly reliable “first-pass” filter to safely eliminate more than half of irrelevant literature without missing critical studies, allowing resources to be reallocated to full-text review and data extraction.

1 Introduction

Systematic Reviews (SRs) are considered one of the best ways to answer a research question; however, they are time-consuming, resource heavy and can be out of date by the time they are published. They require multiple, complex tasks to produce [1]. Screening literature search results, one of the first tasks in a SR, often consists of reviewing thousands of irrelevant references [2], which must be manually screened by at least two reviewers [3]. This process is often time consuming; Bramer et al. [4] estimated a rate of 60-300 references per hour. Variation in screening speed can be attributed to team capacity, the inclusion and exclusion criteria, and the software use.

Automation tools can assist to reduce this burden by estimating the relevance of references for a given review, providing researchers with a metric to quickly ascertain whether a study is likely or unlikely to meet inclusion criteria. Large Language Models (LLMs) appear to be a promising method to achieve this, however recall continues to be a limiting factor [5]. Recent evaluations of LLMs in literature screening indicate that while average pooled recall frequently exceeds 0.90, the highest achieved recall to date has reached 1.00 (100%) under specific settings utilizing ensemble models and probability calibration, albeit frequently at the cost of reduced precision [6]. To effectively automate screening, recall (the number of relevant studies included) needs to be near 1.00 (meaning all relevant studies were correctly identified). To date no LLM has been able to achieve a recall of 1.00 consistently across reviews, while reducing the screening burden in a measurable way [7].

To address the need of reducing the screening burden, we developed the MechaScreener. The MechaScreener is an LLM screening tool available via the Evidence Review Accelerator (TERA) [8]. Unlike traditional machine learning tools that require reviewers to manually screen a subset of references to train the algorithm (e.g., active learning models such as Abstrackr, ASReview, Rayyan and others [9]), MechaScreener is a “zero-shot” tool. This means it does not require any training examples. Instead, it relies on the LLM’s pre-existing language capabilities to evaluate references; users simply upload their references and enter their review’s eligibility criteria in plain text. This study outlines the development of the MechaScreener and its screening results across 15 systematic reviews.

2 Methods

2.1 System Architecture

MechaScreener development started in 2024 using a Node.js/TypeScript backend environment with a Vue 3 frontend. The system processes reference citations in batches, extracting the title and abstract and pairing them with a user-provided screening criteria string; formatted as Population, Intervention, Comparator, Outcome, Type of study (PICOT) criteria [10] or general inclusion/exclusion rules).

2.2 Prompting and Scoring

The MechaScreener system utilises a structured prompt consisting of:

1. **System Prompt:** Defined instructions for the LLM to act as a screening assistant.
2. **Reference Data:** Title and abstract of the paper.
3. **Criteria:** The eligibility criteria for the specific review.

In the context of Large Language Models, **temperature** is a parameter that controls the randomness or creativity of the model’s output distribution, while **top_k** restricts the model to sample only from the k most probable next tokens. To ensure the screening process is completely deterministic and highly reproducible for research purposes, we utilised a model configuration with a **temperature** of 0 and **top_k** of 1, effectively forcing the model to always select the single most likely response.

The model is instructed to provide a score from 1 (low probability of inclusion) to 5 (high probability of inclusion).

2.3 Experimental Design

The evaluation of MechaScreener was conducted in two phases:

- **Development Dataset:** Five control reference libraries were used to calibrate the prompt and scoring threshold to maximise recall and specificity.
- **Evaluation Dataset:** Ten additional reference libraries were used for independent validation of the tool’s performance across different medical and social science domains (e.g., COVID-19, diabetes, bone health). This includes five SRs that were looking for Randomised Controlled Trials (RCTs) and five SRs that were looking at non-RCT studies.

All 15 reference libraries are available online in our GitHub repository. The eligibility criteria prompts provided to the LLM were developed by three individuals with varying levels of systematic review experience (a novice reviewer, a PhD student, and an experienced reviewer), with each individual designing the prompts for five libraries. The exact eligibility criteria provided to the LLM for each library are detailed in Supplementary File 1. This file can also be used as a prompt guide for using MechaScreener. The 10 evaluation studies were sourced from previous Cochrane reviews. As title/abstract screening results are not available, we used the final full-text inclusion decisions to determine what the relevant studies were.

To compile the set of reviews used in the evaluation we reran searches from 10 randomly selected Cochrane reviews. This was done by searching PubMed for a Cochrane review, then generating a random number using the Google random number generator. This number was between one and the total number of search results found, e.g., if 1000 results were found the random number was between 1 and 1000. The search result that corresponded to this number was checked to see if it met our inclusion criteria of 1) having searched three or more databases; 2) reported search strategies for all databases; 3) have an identifiable list of included studies; 4) have a minimum of 6 to a maximum of 50 included studies; and for five of the reviews including randomised controlled trials and for the other five including study types other than randomised controlled trials. Any references missing abstract data were excluded from the evaluation.

2.4 Metrics

The performance of MechaScreener is evaluated primarily using two metrics: recall (sensitivity) and specificity (selectivity).

Recall (or sensitivity) measures the proportion of actual relevant studies that were correctly identified by the tool. In the context of systematic reviews, maximising recall is paramount to ensure no critical literature is missed. It is defined as:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (1)$$

Specificity (or selectivity) measures the proportion of irrelevant studies that were correctly excluded. A higher specificity indicates a greater reduction in the manual screening workload, as fewer irrelevant articles are passed to the human reviewers. It is defined as:

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}} \quad (2)$$

Wilson Score Confidence Interval (95% CI) is used to calculate the confidence bounds for recall and specificity. Standard Wald confidence intervals fail mathematically when proportions reach 0% or 100% (as seen with perfect recall). The Wilson Score Interval is robust for these extreme values [11]. It is calculated as:

$$\text{CI} = \frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \quad (3)$$

where $\hat{p}$ is the observed proportion, n is the total number of observations, and z is the z-score (1.96 for a 95% confidence level).

2.5 Development Results

Five control libraries were used for development to maximise recall (number of references correctly included) and specificity (number of references correctly excluded). These libraries were compiled based on previous reviews conducted at the Institute for Evidence-Based Healthcare (IEBH). The developmental results for these libraries are presented in Table 1.

Table 1: Results Across Five Development Libraries

Library Name	No. of Refs	Include Studies Correctly Included	Exclude Studies Correctly Excluded
Antibiotic Prescribing and Tele-health [12]	934	13/13 (100%)	612/921 (66%)
Long Covid [13]	1,216	16/16 (100%)	552/1,200 (46%)
Natural History Primary Care [14]	5,913	11/11 (100%)	3,169/5,902 (54%)
Non Drug Interventions [15]	5,342	35/35 (100%)	3,972/5,307 (75%)
Salt Substitution [16]	889	23/23 (100%)	733/866 (85%)

2.6 Evaluation Dataset

Ten reference libraries were used for evaluation of performance of MechaScreener. These 10 libraries include a diverse range of topics shown below (five studies that were RCTs and five studies that were non-RCTs).

Table 2: Overview of References and Missing Abstracts in Evaluation Libraries

Library Name	Total References	Missing Abstracts
RCT Studies		
Balneotherapy for Chronic Venous [17]	338	11
Calcium Vitamin D for Bones [18]	5,756	255
Methylxanthine for Apnea [19]	2,781	334
Phosphodiesterase 5 Inhibitors [20]	2,672	338
Topical and Oral Steroids for Om [21]	4,292	583
Non-RCT Studies		
COVID-19 Measures in Schools [22]	2,162	21
Diabetes as a TB Risk Factor [23]	5,177	165
Falls Interventions [24]	18,098	1,095
Topical Fluoride causing Dental Fluorosis [25]	11,829	2,527
Sanitation for Prevention Diarrhoea [26]	10,770	297

3 Results

MechaScreener was tested across 10 evaluation libraries containing a total of over 58,000 references. Figure 1 summarises the performance metrics visually across each library and highlights the overall average for the evaluation set.

Within the evaluation dataset, the system achieved a perfect mean recall score (1.000), meaning no relevant articles were missed by the tool. The mean specificity was 0.609, indicating that MechaScreener correctly excluded 61.0% of irrelevant articles, significantly narrowing the pool for human review.

An individual breakdown of each evaluation library is presented in Table 3.

Table 3: Results Across 10 Evaluation Libraries

Library Name	No. of Refs	Include Studies Correctly Included	Exclude Studies Correctly Excluded
Balneotherapy for Chronic Venous	327	9/9 (100%)	259/318 (81%)
Calcium Vitamin D for Bones	5,493	6/6 (100%)	1,825/5,487 (33%)
Methylxanthine for Apnea	2,446	28/28 (100%)	1,422/2,418 (59%)
Phosphodiesterase 5 Inhibitors	2,333	6/6 (100%)	2,128/2,327 (91%)
Topical and Oral Steroids for Om	3,706	28/28 (100%)	3,110/3,678 (85%)
COVID-19 Measures in Schools	2,141	30/30 (100%)	439/2,111 (21%)
Diabetes as a TB Risk Factor	5,011	48/48 (100%)	1,948/4,963 (39%)
Falls Interventions	17,000	9/9 (100%)	8,102/16,991 (48%)
Topical Fluoride causing Dental Fluorosis	9,244	40/40 (100%)	6,357/9,204 (69%)
Sanitation for Prevention Diarrhoea	10,473	31/31 (100%)	8,684/10,442 (83%)

Furthermore, we calculated the 95% Wilson Score Confidence Intervals. Table 4 presents these statistical measurements, alongside the macro-average (mean of proportions) and micro-average (pooled data) across all evaluation datasets.

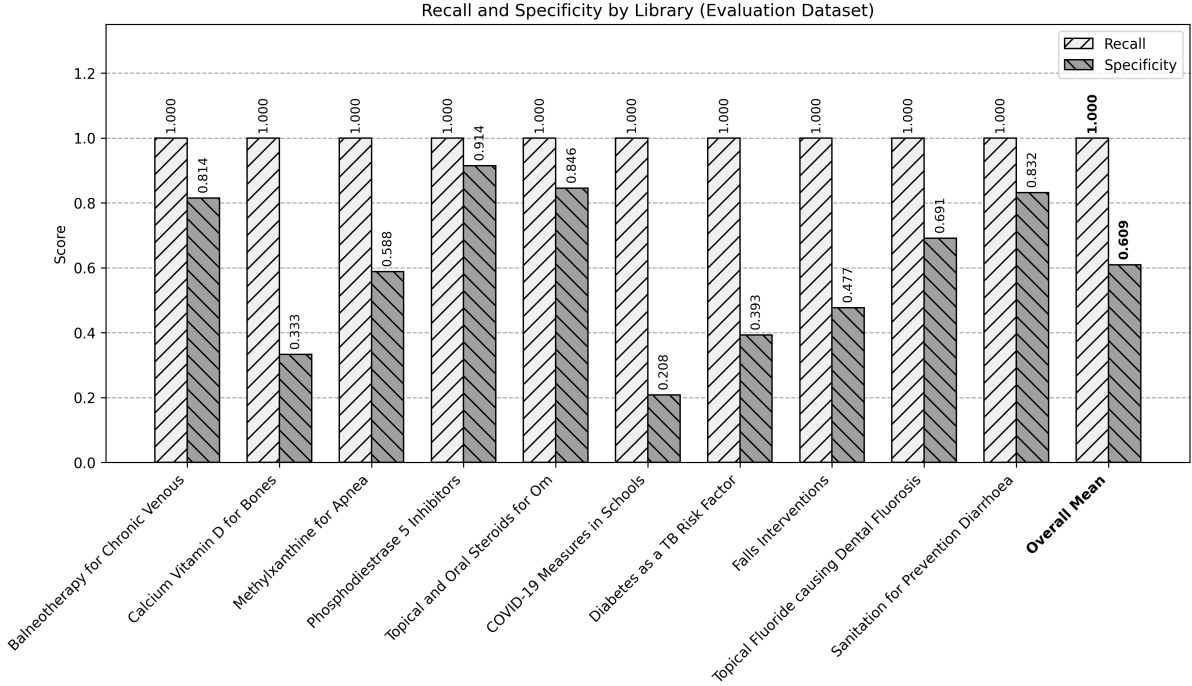


Figure 1: Recall and Specificity Across 10 Evaluation Libraries, Including the Overall Mean.

4 Discussion

In the context of SRs, recall is universally recognised as the most critical metric; inappropriately excluding a single relevant study can introduce bias and potentially invalidate the findings of the entire review. Our results demonstrate that MechaScreeners, when prompted with a structured 1–5 scoring system, can act as a highly effective and safe “first-pass” filter.

The primary value of MechaScreeners lies in its ability to consistently maintain a perfect mean recall (1.000) across all evaluated libraries while safely removing an average of approximately 61% of the irrelevant literature (mean specificity of 0.609). This translates to a substantial reduction in the manual screening workload for researchers, reallocating valuable time and human resources towards downstream tasks such as full-text review, data extraction, and risk of bias assessment.

4.1 Performance Variance and Specificity

While the recall remained stable at 100% across both the development and evaluation datasets, we observed notable variance in specificity depending on the systematic review topic and study design. This variance demonstrates that the LLM’s capacity to confidently discard irrelevant literature is highly dependent on the precision of the research question and the standardisation of the field’s terminology.

For instance, highly specific pharmacological interventions such as *Phosphodiesterase 5 Inhibitors* [20] achieved a specificity of 0.914, effectively eliminating over 90% of irrelevant studies. Clinical trials in these domains typically feature highly structured abstracts and unambiguous PICOT elements, allowing the LLM to easily detect off-target papers.

Conversely, broader or more complex topics, particularly those involving non-randomised controlled trials (non-RCTs) like *Covid COVID-19 Measures in Schools* [22] or *Diabetes as a TB Risk Factor* [23], exhibited lower specificities (0.208 and 0.393, respectively). These public health and observational fields often involve diverse study designs, varied outcome measures, and less structured reporting. When abstract information is vague, lacks explicit PICOT details, or touches on related thematic concepts without matching precise criteria, MechaScreeners

Table 4: Statistical Summary Across 10 Evaluation Libraries

Library Name	Recall (95% CI)	Specificity (95% CI)
Balneotherapy for Chronic Venous	1.000 (0.701-1.000)	0.814 (0.768-0.853)
Calcium Vitamin D for Bones	1.000 (0.610-1.000)	0.333 (0.320-0.345)
Methylxanthine for Apnea	1.000 (0.879-1.000)	0.588 (0.568-0.608)
Phosphodiesterase 5 Inhibitors	1.000 (0.610-1.000)	0.914 (0.902-0.925)
Topical and Oral Steroids for Om	1.000 (0.879-1.000)	0.846 (0.834-0.857)
COVID-19 Measures in Schools	1.000 (0.886-1.000)	0.208 (0.191-0.226)
Diabetes as a TB Risk Factor	1.000 (0.926-1.000)	0.393 (0.379-0.406)
Falls Interventions	1.000 (0.701-1.000)	0.477 (0.469-0.484)
Topical Fluoride causing Dental Fluorosis	1.000 (0.912-1.000)	0.691 (0.681-0.700)
Sanitation for Prevention Diarrhoea	1.000 (0.890-1.000)	0.832 (0.824-0.839)
Macro-Average (Mean)	1.000	0.609
Pooled (Micro-Average)	1.000 (0.984-1.000)	0.592 (0.588-0.596)

conservatively assigns a higher probability of inclusion. This built-in conservatism prevents the accidental exclusion of relevant studies, correctly prioritising a perfect recall over maximal workload reduction when faced with ambiguity.

4.2 Limitations

Despite the robust performance, several limitations must be acknowledged. First, as noted in the methodology, the evaluation dataset’s ground truth for ‘relevant’ studies was based on final full-text inclusion decisions rather than initial human title and abstract screening decisions. Because abstracts often lack complete methodological details, an automated title and abstract screener may correctly flag a study as potentially relevant based on the abstract alone, only for it to be counted as a false positive because the study was ultimately excluded during the human full-text review phase. Consequently, the reported specificity may artificially underestimate the tool’s true precision at the abstract screening level.

Second, the current methodology relies heavily on the availability of study abstracts. As highlighted in Table 2, thousands of references across the tested libraries (e.g. over 2,500 in the *Topical Fluoride causing Dental Fluorosis* library) lacked abstract data. Studies without abstracts are inherently difficult to screen automatically and often default to human review, reducing the overall workload savings.

Third, MechaScreeners does not replace human screening. It functions strictly as a “first-pass” filter, meaning that any results the tool flags as potentially relevant still need to be manually screened by a human reviewer to confirm eligibility. Furthermore, MechaScreeners is designed explicitly as a title and abstract screening assistant and does not perform full-text screening, meaning human reviewers must also assess the remaining literature in full detail. Finally, while we utilised a deterministic approach (i.e. `temperature = 0`) to ensure reproducibility, the system’s performance remains fundamentally tied to the quality, clarity, and comprehensiveness of the user-provided eligibility criteria. Poorly defined criteria may lead to lower specificity as the model struggles to differentiate borderline studies.

4.3 Future Directions

Future research should explore the integration of MechaScreeners directly into standard systematic review workflows and reference management platforms. Evaluating its performance across a broader range of foundational LLMs, as well as refining the prompt architecture to better

handle title-only references, are key next steps. Furthermore, expanding the system’s capabilities to assist with automated full-text screening and data extraction could further streamline the evidence synthesis pipeline. Overall, these findings solidify the role of LLMs as powerful, dependable augmentations to traditional systematic review methodologies.

5 Conclusion

MechaScreener provides a reliable method for screening research articles. By leveraging large-scale LLMs and a probabilistic scoring approach, researchers can use MechaScreener to reduce their screening workload by more than half while maintaining the rigorous standards required for systematic literature synthesis.

Declarations

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Competing interests: The authors of this study are responsible for the development of MechaScreener and hence may present bias towards favourable findings. However, we encourage independent testing of the method and have made the code and testing datasets open-source and publicly available to be as transparent as possible and improve replicability.

Data availability statement: The dataset (relevance scores generated by Mechascreener and ground truth include/exclude decisions), source code for the analysis, along with the scripts used to generate the figures and tables presented in this paper, are publicly archived and available in the GitHub repository: <https://github.com/IEBH/mechascreener-paper>.

Ethics approval statement: Not applicable. This study involves the methodological analysis of previously published literature and software performance, and does not involve human or animal subjects.

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